

Shubham Shah

AI/ML Engineer — LLM Agents & RAG Systems

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Profile

AI systems-focused undergraduate building production-grade LLM agents and retrieval systems for real-world workflows. Experienced in designing multi-step agent pipelines, integrating AI with APIs and data systems, and deploying scalable solutions. Strong interest in applying AI to optimize business processes, automate decision-making, and improve operational efficiency.

Technical Skills

Agent & LLM Systems: Autonomous Agents, Multi-step Workflows, RAG Pipelines, Prompt Engineering, Tool Use, Guardrails

Machine Learning: Transformers, Embeddings, XGBoost, Semantic Search

Data Systems: Document Parsing, Chunking, Knowledge Indexing, Vector Search (FAISS), TF-IDF

Evaluation: Precision, Recall, F1-score, Benchmarking, Error Analysis

Programming: Python, Java, SQL

Frameworks/Tools: FastAPI, Spring Boot, scikit-learn, Pandas, NumPy, Docker, Git, Linux

Systems: REST APIs, Async Pipelines, Workflow Orchestration, Cloud Deployment (Render)

Education

BITS Pilani Digital + Scaler School of Technology

2024 – Present

B.Sc. in Computer Science Engineering + M.Sc. (Woolf University)

CGPA: 9.6 / 10.0

Projects

PDF Grounded Chatbot (PDF RAG + Citations + Guardrails)

GitHub — Live

- Built a PDF-constrained Streamlit chatbot that answers questions strictly from uploaded PDFs with page-number citations.
- Implemented local PDF parsing with PyMuPDF and lexical retrieval to supply only relevant page excerpts to the OpenAI Responses API.
- Designed grounding guardrails to refuse unsupported queries with a fixed out-of-scope response and reject uncited answers.
- Added conservative extractive fallback and multilingual support while preserving citation reliability across valid PDF-based queries.

CommentAI — Autonomous AI Agent (LLM + Workflow Automation + Monitoring)

GitHub

- Built a fully autonomous AI agent that engages with Instagram and LinkedIn posts using a locally hosted LLM (Llama-3 via Ollama).
- Designed end-to-end workflow system for content discovery, decision-making, generation, and execution.
- Processed comments **10+ posts/day** with automated scheduling, human-like delays, and execution pipelines.
- Implemented advanced guardrails (duplicate prevention, creator cooldowns, semantic filtering), reducing spam/irrelevant outputs by **40%**.
- Developed real-time FastAPI + React dashboard to monitor agent activity, engagement metrics, and system behavior.
- Eliminated API dependency via local inference, reducing operational cost to **zero** and improving control over deployment.

CCPA Compliance AI System (Enterprise Legal Reasoning + RAG)

GitHub

- Built AI system to analyze business workflows and detect regulatory violations using legal document retrieval.
- Indexed **10k+ legal text chunks** and improved retrieval precision by **20%**.
- Designed constrained reasoning pipeline reducing unsupported outputs by **30%**.
- Structured outputs for reliable downstream decision-making and compliance evaluation.

Razorpay E-commerce Backend (Systems + Payments + Workflows)

GitHub

- Built backend supporting concurrent users with end-to-end order and payment workflows.
- Integrated Razorpay APIs with secure verification and webhook-driven processing.
- Designed transactional systems ensuring reliability in order lifecycle and failure handling.